

Pilot Simulation and Data-Informed Evaluation

Story of Implementation

Due to limited access to the intended learner population, I could not conduct a full live pilot. Instead, I used a structured pilot simulation paired with evidence-informed, design-based evaluation to assess the instructional solution before broader implementation. This approach allowed me to evaluate the flow of instruction, clarity, pacing, and alignment to learner needs, which is consistent with instructional design practices when access is limited (Gustafson & Branch, 2007; Branch, 2018).

I reviewed the instructional solution on April 24, 2026, during a single session lasting about 45–50 minutes. The review took place in the virtual, asynchronous format the lesson was designed for, so I could experience it the same way learners would—independently move through the materials and activities.

Even though I did not have a live learner present, I kept the target learner in focus throughout the evaluation—high school students preparing for postsecondary decisions who may not have much experience independently comparing career options. Based on research, learners at this stage benefit from clear structure, explicit guidance, and manageable cognitive load when working through complex decision-making tasks (Hmelo-Silver et al., 2007; Sweller et al., 2019). That understanding guided how I evaluated clarity, sequencing, and pacing.

Role During Implementation

During the simulation, I took on the role of both **instructional designer and evaluator**. I intentionally worked through the lesson from the learner’s perspective, step by step, paying attention to how each part functioned as designed.

As I moved through the instructional sequence, I documented where:

- directions were clear or unclear
- transitions felt smooth or abrupt
- pacing supported or challenged understanding
- cognitive demand increased

This helped me identify what was working well and where the design itself might create unnecessary difficulty. This type of evaluation aligns with ADDIE, where instructional materials are reviewed and refined before full implementation to reduce barriers and improve effectiveness (Gustafson & Branch, 2007; Branch, 2018).

Effectiveness of the Instructional Design

What Worked Well

Overall, the design shows strong alignment between the objectives, activities, and assessment. The lesson flows in a way that makes sense—starting with self-assessment, moving into exploration, then reflection, and finally application. That progression supports how learners build understanding over time, especially for something like career decision-making.

The self-assessment activity is placed at the beginning on purpose, and it works. It helps activate prior thinking and makes the learning feel relevant right away, which supports engagement (Clark & Mayer, 2016).

The reflection and comparison activities also do what they're supposed to do—they push learners to think, not just recall information. Learners must analyze options and make decisions, which aligns with higher-order thinking and supports independence (Anderson & Krathwohl, 2001).

What Needs Improvement

During the review, I noticed that some instructions had to be reread to fully understand what was expected, especially in the career comparison activity, where multiple criteria (salary, training, time, personal fit) were introduced at once.

That's a signal of increased cognitive load. For learners who are not used to processing multiple pieces of information at the same time, this can make the task feel more difficult than it needs to be.

Research supports this—when too many new elements are presented at once without chunking or clear guidance, it can reduce comprehension (Sweller et al., 2019).

Also, when I went through the lesson as one continuous session, it required a lot of sustained attention across multiple demanding tasks. That suggests the pacing could be improved by breaking the lesson into smaller, more manageable sections.

Formative Evaluation Questions

To support future implementation and collect meaningful data, these are the questions I would use:

- Which parts of the lesson were the most confusing or unclear?
- Where did you feel the need to pause or reread directions?

- Which activity helped you the most when comparing your career options?
- After completing the lesson, how confident do you feel making a postsecondary decision?

These questions are intentional—they focus on specific parts of the design instead of general feedback, which makes revisions more targeted (Black & Wiliam, 2009).

Future Modifications

Based on what I found during the evaluation, there are a few clear next steps.

First, instructional directions, especially in comparison activity—need to be simplified. Adding short examples will help clarify expectations and reduce the need for rereading.

Second, the lesson should be broken into smaller sections with clearer transitions. This will help manage cognitive load and make it easier for learners to stay engaged throughout.

These changes are directly tied to what I observed during the evaluation and are supported by instructional design principles around clarity, scaffolding, and chunking.

Using this type of feedback to revise the lesson reflects the iterative nature of ADDIE, where instruction is continuously improved before full implementation (Branch, 2018; Molenda, 2015).

References

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